universe right now—if you know where *every* particle is and how fast and in what direction each is moving—then, Newton and Einstein agree, you can, in principle, use the laws of physics to predict everything about the universe arbitrarily far into the future or to figure out what it was like arbitrarily far into the past.[2]

Quantum mechanics breaks with this tradition. We *can't* ever know the exact location and exact velocity of even a single particle. We *can't* predict with total certainty the outcome of even the simplest of experiments, let alone the evolution of the entire cosmos. Quantum mechanics shows that the best we can ever do is predict the *probability* that an experiment will turn out this way or that. And as quantum mechanics has been verified through decades of fantastically accurate experiments, the Newtonian cosmic clock, even with its Einsteinian updating, is an untenable metaphor; it is demonstrably *not* how the world works.

But the break with the past is yet more complete. Even though Newton's and Einstein's theories differ sharply on the nature of space and time, they do agree on certain basic facts, certain truths that appear to be self-evident. If there is space between two objects—if there are two birds in the sky and one is way off to your right and the other is way off to your left—we can and do consider the two objects to be independent. We regard them as separate and distinct entities. Space, whatever it is fundamentally, provides the medium that separates and distinguishes one object from another. That is what space does. Things occupying different locations in space are different things. Moreover, in order for one object to influence another, it must in some way negotiate the space that separates them. One bird can fly to the other, traversing the space between them, and then peck or nudge its companion. One person can influence another by shooting a slingshot, causing a pebble to traverse the space between them, or by yelling, causing a domino effect of bouncing air molecules, one jostling the next until some bang into the recipient's eardrum. Being yet more sophisticated, one can exert influence on another by firing a laser, causing an electromagnetic wave —a beam of light—to traverse the intervening space; or, being more ambitious (like the extraterrestrial pranksters of last chapter) one can shake or move a massive body (like the moon) sending a gravitational disturbance speeding from one location to another. To be sure, if we are